Certifying Cumulative Harm Across Training Checkpoints

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Abstract

A later training stage can erase behavior learned during safety tuning. We compute the nearest update satisfying one first-order harm constraint, then show why that update alone can still raise harm. A curvature term yields a sufficient finite-step condition. For repeated updates along a fixed search line, we give the sharp cumulative harm interval from a Lipschitz bound and interval observations of the directional derivative. An exact oracle checks the bounds and distinguishes preserved harm allowance from a rule that resets its allowance after each step. For checkpoints on a turning path, exact gradients and a global smoothness bound give sharp endpoint values through shortest paths. A source-aware search can certify a cap when a bounded number of intermediate measurements may be faulty.

Summary and scope

Question. How much can a protected harm score increase across a sequence of tentative updates?

Finding. On a fixed search line, interval observations of a Lipschitz directional derivative give exact endpoints for cumulative change. Exact gradient vectors give a separate certificate that can link nonadjacent checkpoints. One-sided projection or a renewed allowance at every step can miss a breach.

Evidence and scope. The interval is proved and checked by a lattice oracle. A tuning team can certify each tentative release against the same starting value. The guarantee applies to the specified surrogate and a certified curvature bound.

1 A protected behavioral objective

Suppose $H(w)$ is a differentiable harm surrogate on a frozen training slice. It might score whether a tool-use policy acts after a veto, with the action validator separated from the reward model. Let $a = \nabla H(w) \neq 0$, and let g be the update proposed by the next training objective. Gradient surgery uses projections to reduce gradient conflict in multitask learning [1]. Here a designated surrogate defines a one-sided guard for a proposed update.

Proposition 1. *The Euclidean-nearest update with $a^\top d \leq 0$ is*

$$d = g - \frac{\max(0, a^\top g)}{\|a\|^2} a.$$

If ∇H is L-Lipschitz on the update segment, then $H(w+d) \leq H(w) + a^\top d + (L/2)\|d\|^2$.

Proof. When g is feasible, its distance to itself is zero. Otherwise write g and any feasible vector in normal and tangential coordinates. The tangential component can match g, while the closest feasible normal coordinate is zero. The formula follows. Integrating $\nabla H(w+td)^\top d$ for $0 \leq t \leq 1$ and bounding its difference from $a^\top d$ by $Lt\|d\|^2$ gives the finite-step inequality. □

2 Why a projected step can fail

Take $H(x, y) = x + y^2$ at $(0, 0)$, with $a = (1, 0)$ and $g = (1, 1)$. Projection gives $(0, 1)$, which satisfies the linear guard with equality. Its harm rises from zero to one. If the permitted slack is $m = H_{\max} - H(w)$, a sufficient update condition is $a^\top d + (L/2)\|d\|^2 \le m$. With zero slack and no first-order decrease, a finite step cannot be certified by this inequality when $L > 0$.

3 A sharp cumulative bound on a fixed search line

Let $w(x) = w_0 + xv$ with $\|v\| = 1$, and write $g(x) = \frac{d}{dx} H(w(x))$. Suppose g is L-Lipschitz on $[0, T]$ and that certified observations restrict $g(x_i) \in [\ell_i, u_i]$ at finitely many nodes, including 0 and T. A tentative checkpoint can supply the last observation before an update is accepted. Define the upper and lower extensions

$$U(x) = \min_i \{u_i + L|x - x_i|\}, \qquad V(x) = \max_i \{\ell_i - L|x - x_i|\}.$$

These are the interval-observation version of the extremal McShane–Whitney envelopes for Lipschitz functions [2].

Proposition 2. *The observations are feasible if and only if $U(x_i) \ge \ell_i$ for every node. When feasible, the sharp interval for $H(w(T)) - H(w(0))$ is*

$$\left[\int_0^T V(x)\,dx,\ \int_0^T U(x)\,dx\right].$$

Proof. Every feasible g lies between V and U by the Lipschitz bound. Conversely a finite minimum or maximum of L-Lipschitz functions is L-Lipschitz. At a node, $U(x_i) \le u_i$, and the feasibility condition gives the other inequality. That condition also states $\ell_i \le u_j + L|x_i - x_j|$ for all node pairs. Reversing the roles of the indices shows $V(x_i) \le u_i$, while the term for node i gives $V(x_i) \ge \ell_i$. Both envelopes are admissible derivatives. Integrating each gives a witness with the same initial value. Convex combinations of the two derivatives attain intermediate changes. □

Between neighboring nodes at distance d, let a and b be the tightened values of U at the left and right nodes. All cones from the left have slope L inside this interval and all cones from the right have slope $-L$. Thus the upper envelope at distance t from the left is $\min\{a + Lt, b + L(d - t)\}$. For $L > 0$ its integral is

$$\frac{d(a+b)}{2} + \frac{Ld^2}{4} - \frac{(b-a)^2}{4L}.$$

The formula follows by splitting the integral at $t = (b - a + Ld)/(2L)$, which lies in $[0, d]$ by the Lipschitz condition. For $L = 0$ the feasible derivative is constant between the nodes and the integral is da. Apply the same upper calculation to $-g$ to get the lower endpoint. With zero endpoint derivatives on a segment of length two and $L = 1$, the allowed increase can still reach one.

Keep the harm cap anchored to the protected reference checkpoint. Certify the cumulative increase at every checkpoint proposed for release. For example, with $H(w(x)) = x$, two increments of $3/5$ each fit under a freshly reset allowance of one but their cumulative increase is $6/5$. If the search turns to a different line, valid per-segment upper bounds can be added from the same reference value. The resulting bound need not be sharp on a path with changing directions.

4 Linking checkpoints on a turning path

Suppose the full gradients $g_i = \nabla H(x_i)$ are known at finitely many checkpoints x_i. Assume $L > 0$ and that the gradient of H is L-Lipschitz on the whole parameter space. This stronger domain assumption lets the certificate use chords between checkpoints as well as the proposed update segments. Write $d_{ij} = x_j - x_i$ and set

$$c_{ij} = \frac{1}{2}(g_i + g_j)^\top d_{ij} + \frac{L}{4}\|d_{ij}\|^2 - \frac{1}{4L}\|g_j - g_i\|^2.$$

Proposition 3. *The checkpoint gradients can come from a common globally L-smooth surrogate if and only if the complete directed graph with weights c_{ij} has no negative cycle. In that case the sharp interval for the change from checkpoint 0 to checkpoint T is*

$$\big[-\operatorname{dist}(T, 0),\ \operatorname{dist}(0, T)\big].$$

Proof. Let h_i be the unknown checkpoint values. Add the quadratic $L\|x\|^2/2$ to H to obtain a convex function F with $2L$-Lipschitz gradient. Conversely, subtracting that quadratic from any convex $2L$-smooth function gives an L-Lipschitz gradient. To see this, for a displacement d and gradient difference ΔG of F, the smooth convex gradient inequality gives $\langle \Delta G, d\rangle \geq \|\Delta G\|^2/(2L)$. Expanding $\|\Delta G - Ld\|^2$ bounds it by $L^2\|d\|^2$.

The smooth convex interpolation theorem of Taylor, Hendrickx, and Glineur [3] requires $F_i \geq F_j + \nabla F_j^\top (x_i - x_j) + \|\nabla F_i - \nabla F_j\|^2/(4L)$ for every ordered pair. Substituting the shifted values and gradients reduces this to $h_j - h_i \leq c_{ij}$. Thus these difference inequalities are necessary and sufficient for interpolation.

Summing around a cycle rules out a negative cycle. Conversely, with no negative cycle, distances from any root r give feasible values $h_i = \operatorname{dist}(r, i)$ by the triangle inequality for shortest paths. Summing along paths from 0 to T bounds the increase above by $\operatorname{dist}(0, T)$. The root $r = 0$ attains this bound, and $r = T$ attains the lower bound. Add a constant to match the protected reference value. Interpolating between the feasible value vectors gives all intermediate changes. □

For $L = 1$, take $(0,0), (1,0), (0,1)$ in that order, with zero gradient at each. The consecutive segment bounds sum to $3/4$. The direct edge bounds the sharp increase by $1/4$. Gradient consistency also requires more than the pairwise Lipschitz check. At the same checkpoints, set the gradients to $(0,0), (0,1), (-1,0)$. The two-edge cycles all have weight zero, but the reverse triangle has weight -1. These vectors cannot be gradients of the stipulated surrogate.

On 128 finite fixtures, graph closure agreed with direct enumeration of simple paths and cycles. All 64 gradients generated from smooth quadratics were feasible, and the known values met every certified bound. The 64 unconstrained gradient records were rejected. In 32 feasible fixtures the graph improved on summing consecutive bounds. This is a finite exact check of the graph calculation. It supplies no measured global smoothness constant for a real model.

For gradient uncertainty sets, retain the vector g_i as a decision variable in each set and keep the original difference inequalities. With convex uncertainty sets these are convex constraints after moving the squared gradient difference to the left side. This gives a finite formulation for a sharp bound under the same smoothness model. The shortest-path algorithm above assumes the gradients are fixed.

4.1 A cap that tolerates faulty gradient sources

Some intermediate gradient measurements can share a data pipeline. Treat each such pipeline as one source and assume that at most k sources can be faulty. A faulty source contributes no trusted

gradients. The endpoint gradients and the global smoothness bound remain trusted. We want the increase from 0 to T to be at most B for every admissible choice of faulty sources and every surrogate that fits the remaining measurements.

Proposition 4. *The following finite search either certifies the cap, finds a compatible violation, or establishes that the measurements are inconsistent within the fault budget. Begin with no sources removed. At an inconsistent node, find a negative cycle and branch on its removable sources. At a consistent node, let P be a shortest path from 0 to T. If its weight exceeds B, report a violation. Otherwise branch on the removable sources along P. Stop a branch when the budget is exhausted or the witness contains no removable source. Certify only if at least one consistent retained graph was found and there is no violation.*

Proof. Deleting a source removes its measurement vertices and leaves all other weights unchanged. Any graph extending an inconsistent node by further deletions must break the negative cycle before it can become consistent. At a consistent node, the distance from 0 to T is attained by a smooth surrogate, so a value above B is a valid counterexample. If the distance is at most B, any graph that keeps the path P still has an upper bound at most B. A violating extension must therefore remove a source on that path. These two witness rules preserve every possible counterexample, and each branch consumes one source. If there is no consistent node, all possible routes to consistency were searched through the negative-cycle branches. This yields the separate inconsistency decision. □

For example, take endpoints $(0, 0), (2, 0)$, $L = 1$, and zero measured gradients. Either intermediate checkpoint $(1, 1/2)$ or $(1, -1/2)$ supplies a path with weight $5/8$. With one allowed fault and distinct intermediate sources, at least one such path survives. With both checkpoints owned by one source, losing that source leaves the direct weight 1, which is attainable. Two measurements help only insofar as the fault model separates their sources. A consistent full graph alone cannot establish tolerance to deletions, since removing observations can increase the distance.

The witness search agreed with exhaustive source-subset enumeration on 1,296 finite cap decisions. Of these, 597 were inconsistent, 353 certified and 346 admitted a violation. These counts describe the verifier's grid. For a tuning workflow, record which checkpoint measurements share an estimator, data slice or custody path before setting the source budget. If the trusted assumptions fail, the certificate supplies no basis for accepting the checkpoint.

5 Integrating the guard into development

During capability tuning, estimate protected gradients on a declared replay slice and compare an ordinary update, a projected update, and a projected update with backtracking. For every tentative checkpoint, test the actual protected harm measures before accepting it. Keep a separate final split from the repeated line search. Log vetoed steps and stop when the objective cannot improve within the allowed region. Multiple protected slices require a common feasible set rather than a sequence of projections assumed to preserve all earlier constraints.

6 Local check and open test

The closed form was checked against every feasible comparison on the declared integer grid for 600 nonzero-normal and step pairs. The proof establishes the continuous solution. The finite checks recomputed 8,800 squared-distance inequalities. For the fixed-line extension, a lattice dynamic program independently recovered the exact endpoints for 352 compatible interval histories and

rejected all 160 incompatible ones. It agreed on 2,112 harm-cap decisions. The sharp bound admitted 521 decisions that the left-endpoint-only certificate could not admit on this fixed grid. A real model's gradient noise and unknown curvature can invalidate a certificate, and H can miss harms. A next experiment compares checkpoint retention under matched optimization budgets, including a case where projection succeeds on replay and fails on a hidden slice.

7 Sequential learning and the cost of retention

We fitted a shared logistic scorer to a protected sign task, then moved to a new task with a conflicting sign direction and a free second feature. The protected arm retained at least 70% of the old sign margin and trained only the free direction in the second stage. For the old feature support this implies a bound on replay logistic loss. The matched unguarded arm updated both directions. Across eight seeds and 3,200 held-out cases per task, old-task accuracy was 0/3,200 unguarded and 3,200/3,200 protected. New-task accuracy was 2,785/3,200 unguarded and 1,579/3,200 protected. All 288 proposed free-direction steps were accepted. This example quantifies a severe retention cost in a deliberately conflicting environment. It tests a restricted-direction guard, with the earlier projection result governing a separate mechanism.

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References

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